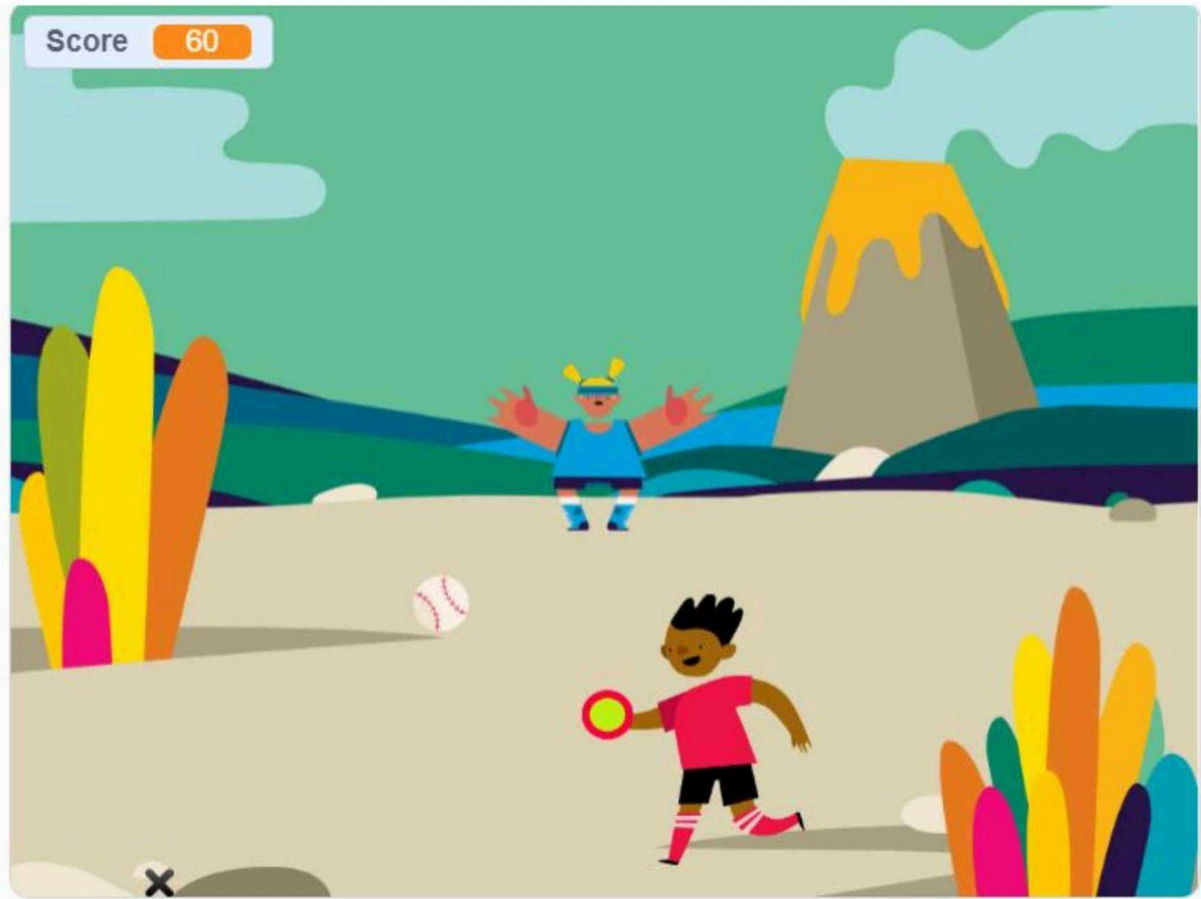


Unit 4

Events & Sensing Workshop

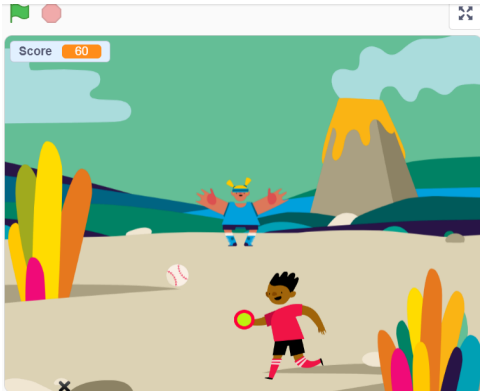


Crazy Catch

Unit 4: Events and Sensing Workshop

Save the file with a new name. Your new name should identify the project.

* Events & Sensing * Crazy Catch * Project 4

Workshop summary The Events menu includes several useful “Hat” blocks. Hat blocks are the only blocks that start a new code stack. Most beginner coders are familiar with the <i>when flag clicked</i> block. This workshop explores several others. In this project, we use Event and Sensing blocks to build a catching game. The player controls the catcher and tries to catch 10 out of 10 pitches from Casey the wild pitcher.		<u>New Terms</u> <u>Interface Terms</u> Sensing blocks, event “hat” blocks <u>Code Block Terms</u> Event, clone, broadcast, message
Class Studio: https://tinyurl.com/StchStudio01 Video of Full Project: n/a This Worksheet:	Starter Project: https://tinyurl.com/StchStarter04 Full Project: https://scratch.mit.edu/projects/664878894	

Mini-Challenges in this Lesson

Click on the mini-challenge link to go directly to that section.

[Mini-Challenge 1: Pitch the Ball](#)

[Mini-Challenge 2: Create Crazy Pitches](#)

[Mini-Challenge 3: Get Ben Moving](#)

[Mini-Challenge 4: Teach Ben to Catch](#)

[Mini-Challenge 5: Keeping Score](#)

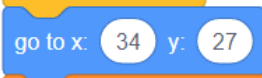
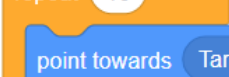
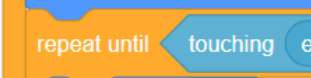
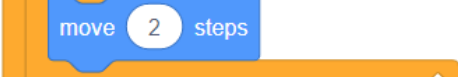
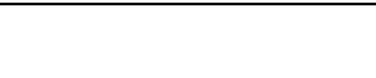
[Mini-Challenge 6: Celebrate Winners and Losers](#)

Mini-Challenge 1: Pitch the Ball

Move the BASEBALL sprite by pointing to the TARGET sprite and moving two steps until the BASEBALL touches the edge.

Procedure

1. The BASEBALL sprite already has some starter code.
2. Add a *repeat 10 times* block.
3. Add a *point towards* block and set it to the TARGET sprite.
4. Add a *repeat until* block inside the other repeat loop.
5. Use the *touching (edge)* diamond block and set it to the edge.
6. Add a *move (___) steps* block.

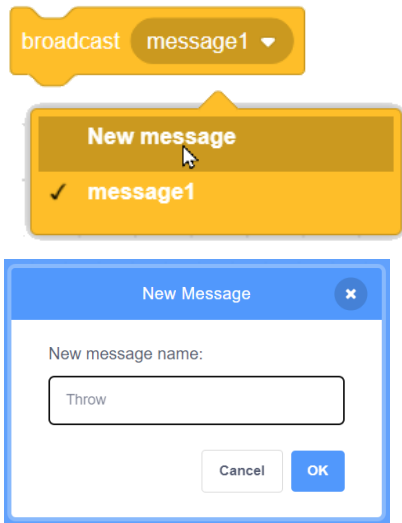
Menu	Code	Code block	Notes
Events	when clicked		Put code stack on BASEBALL sprite.
Motion	go to x: (34) y:(27)		
	Create 2 variables:		The SENSING menu has several diamond blocks.
	Score and Round		
Variables	set Score to 0		We will use 2 touching blocks in this project
Variables	set Round to 0		
Control	repeat (10)		The touching edge diamond will be <i>true</i> whenever the sprite touches an edge.
Motion	point towards (Target)		
Control	repeat until <___>		
Sensing	< touching (edge) >		
Motion	move (2) steps		
Motion	glide (1) second to (Casey)		

Mini-Challenge 2: Create Crazy Pitches

The *broadcast message* block is a special kind of event. It must have a separate *when I receive message* block to work. Because there are two blocks involved, we use broadcast to synchronize the code of separate sprites. The code in the receive block starts running when a message is sent from the broadcast block. We will use messages to signal the TARGET sprite to move to a random location. Then, when the BALL sprite is pitched, it will move to the TARGET's random location.

Procedure

1. Add a *broadcast message* block to the BASEBALL sprite before the *point towards* block. Name the message "Throw."
2. Add a small *wait* block after the *broadcast message* block. The time, .1 seconds works well.

Menu	Code	Code block	Notes
Events	Create a new message Throw		Put code stack on BASEBALL sprite.
Variables • Events • Control • Motion • Control • Sensing • Motion • Motion	set [Round] to (0) repeat (10) times broadcast [Throw] wait (.1) seconds point towards (Target) repeat until < ____ > < touching (edge) > move (2) steps glide (1) second to (Casey)		Put code stack on BASEBALL sprite. Only the bottom half of the code stack is shown. We use a tiny wait to give the Target sprite time to move to its new location. New code is RED .

Procedure

3. Add a *when I receive* block to the Target sprite code page.
 4. Set the message to the one that you created in step #1.
 5. Add a *go to x:() y:()* block. Set y to -175 and leave x blank.
 6. Add a *pick random () to ()* block to the code page.
- Note:** It will not snap to the code stack yet; set the limits to -205 and +205.
7. Drop the *pick random* block into the x value of the *go to x:() y:()* block.

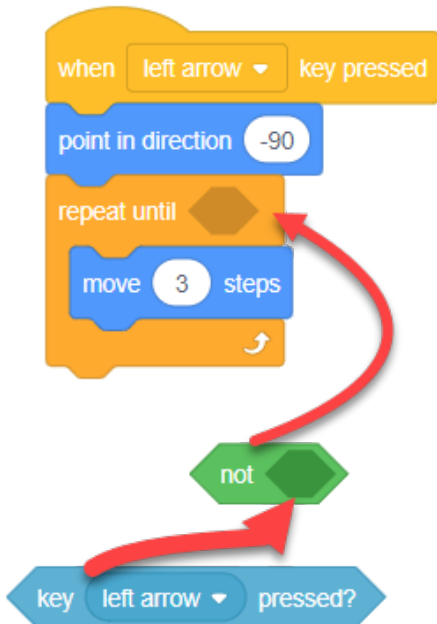
Menu	Code	Code block	Notes
Events	<i>when I receive [Throw]</i>		Put code stack on TARGET sprite.
Motion	<i>go to x:() y: (-175)</i>		
Operators	<i>pick random (-205) (205)</i>		
	It will look like this when the code is finished.		

Mini-Challenge 3: Get Ben Moving

We use the *when key pressed* event to get the BEN sprite moving from side to side. The left arrow key will move BEN left. The right arrow key will move BEN right.

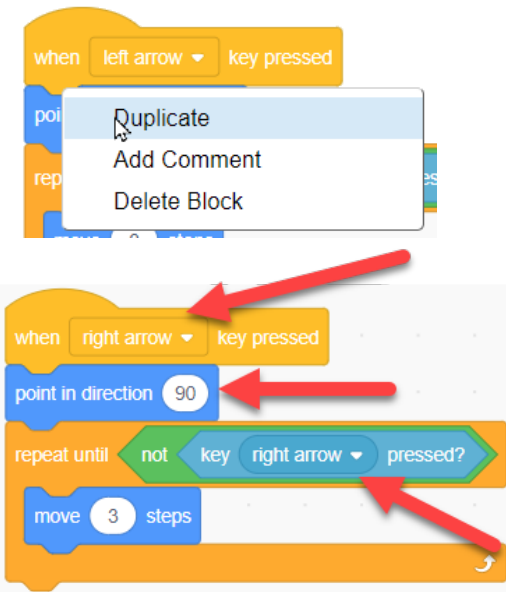
Procedure

1. Add a *when [] key pressed* block to the BEN sprite's code page.
- Note:** Notice it cannot snap to the current code because it is a hat block. Set the dropdown key to "left arrow."
2. Add a *point in direction ()* block and set it for -90. This will make the BEN sprite point left.
 3. Add a *repeat until* block.
 4. Add *move (3) steps* block inside the *repeat until* block.
 5. Add a *not < >* operator block to the *repeat until* diamond.
 6. Add a *key [] pressed* sensing block into the "not" diamond. Set the key value to left arrow.

Menu	Code	Code block	Notes
Events	when [left arrow] key pressed		Put code stack on BEN sprite.
Motion	point in direction (-90)		
Control	repeat until		
Motion	move (3) steps		
Operators	not <___>		
Sensing	key [left arrow] pressed		

Procedure

- Right-click on the *when [left arrow] key pressed* block and select “Duplicate”
- In the copied code stack, change both left arrow dropdowns to right arrow
- In the *point in direction* block, change to 90.

Menu	Code	Code block	Notes
Events	Right-click on the <i>when [left arrow] key pressed</i> and select Duplicate.		Put code stack on BEN sprite.
Motion	change point in direction to 90		
Sensing	change left arrow to right arrow		

Mini-Challenge 4: Teach Ben to Catch

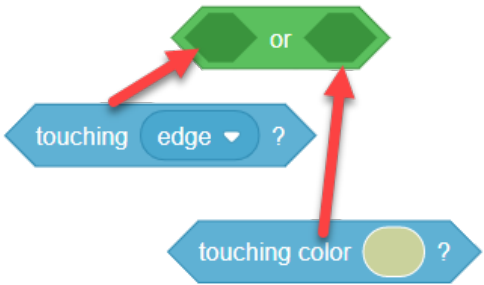
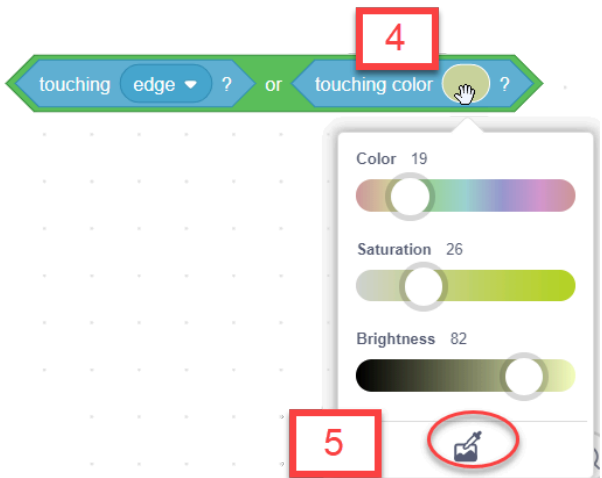
We want BEN to catch the BALL sprite. But the code for him to do it is on the BASEBALL sprite.

We will stop the BASEBALL if it touches the color on the inside of BEN's catcher's mitt.

This will make it seem like BEN has caught the BASEBALL.

Procedure

1. Drop an "or" operator on the BASEBALL's code page.
2. Remove the *touching [edge]* block from the *repeat until* loop and put it on the left side of the "or" block.
3. Drop a *touching color ()* block on the right side of the "or" block.
4. Click on the color panel of the *touching color* diamond. This will open a color menu.
5. Click on the eyedropper at the bottom of the color menu. This activates the color picker.
6. Move your mouse over the stage. It will turn into a magnifier window.
7. Hover over the green part of BEN's catcher's mitt, and the magnifier will turn green.
8. Click on the catcher's mitt to set the color.
9. Drop the completed "or" block back into the repeat until block.

Menu	Code	Code block	Notes
Operators Sensing Sensing	< <_> or <_> touching (edge) touching color ()		Put code stack on BASEBALL sprite.
To change the color inside the circle, use the Color Picker.			When you click on BEN's mitt with the color picker, it will load that color into the "touching color" block.

Menu	Code	Code block	Notes
#7			Put code stack on BASEBALL sprite.
Operators	<__> OR <__>		
Sensing	Touching Edge		
Sensing	Touching Color		

Mini-Challenge 5: Keeping Score

The player wants to know how they are doing. Two variables show her how she's doing: **round** and **score**. The **ROUND** counts how many pitches Casey has thrown. The **score** counts the number of catches. Add two additional blocks *show* and *hide*. This will make the game look less like a tennis match bounce and more like a baseball game toss.

Procedure

1. In the BASEBALL sprite's *repeat 10 times* loop, add a *change [ROUND]* above the *broadcast [throw]* block. This will update the round number.
2. Add a *show* block under the *change [round]* block.
3. At the bottom of the BASEBALL sprite's *repeat 10 times* loop, drop an *if /then* block just above the *glide to Casey* block.
4. Right-click on the *touching color* diamond from Mini-challenge 5 and Duplicate. Then drop the new *touching color* block in the diamond of the *if /then* block.
5. Drop a *change [Score] by 1* block inside the *It Then* block. This updates the score if Ben catches the ball.
6. Add a *Hide* block right above the *Glide* block. This will hide the baseball while it travels back to Casey.

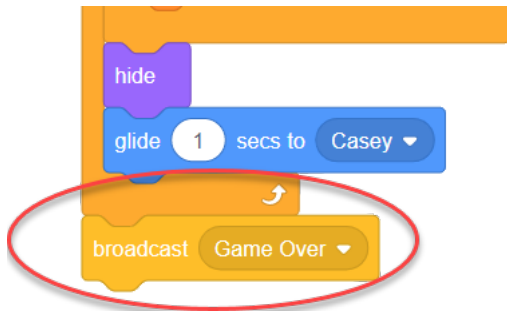
Menu	Code	Code block	Notes
Event Variables Looks Control Sensing Variables Looks	when flag clicked go to x: (); y:() set [score] to (0) set [round] to (0) repeat (10) change [Round] by 1 show broadcast (throw) wait (1) seconds point towards (target) repeat until touching edge ? or touching color ? move 2 steps if touching color ? then change [score] by 1 hide glide (1) sec to (Casey) broadcast (Game Over)		Put code stack on the BASEBALL .

Mini-Challenge 6: Celebrate Winners and Losers

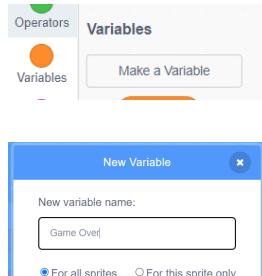
When the game has repeated ten rounds, Casey will stop pitching the ball. That's the time to signal that the game is over. We also need to add two blocks to show and hide the baseball. This will make the game look less like tennis and more like catch.

Procedure

1. In the Baseball sprite, add a new Broadcast block after the end of the Repeat 10 times loop.
This is outside the loop so it doesn't run until the loop has completed all 10 rounds.
2. The Broadcast message is new: Game Over.

Menu	Code	Code block	Notes
Events	broadcast [Game Over]		Put code stack on the BASEBALL .

3. Switch to the Casey sprite
4. Make a new variable: **Game Over**
5. Add a *when I receive [Game Over]* block.
6. Add a *set [Game Over] to ()*. Enter the words **Game Over!** in the second blank.
7. Add a *say () for 2 seconds* block.
8. Drop the **Game Over** variable in the empty say window.
9. Drop two JOIN operator blocks on the canvas.
 - JOIN block 1: Enter “You caught” on the left side.
 - JOIN block 2: Drop the **score** variable on the left side, and type “out of 10” on the right side.
 - Drag the second block into the empty right window of the first.
10. Add another *set [Game Over] to ()* block and drop the Join from #1 into the value.
11. Add a *say(Game Over) for 2 seconds*.

Menu	Code	Code block	Notes
Events	when I receive [Game Over]	when I receive Game Over ▼	Put code stack on CASEY sprite. 
Variables	set [Game Over] to (txt)	set Game Over ▼ to Game Over!	
Looks	say (Game Over) for 2 seconds	say Game Over for 2 seconds	

Variables Operators (2) Variables	set [Game Over] to Pull 2 Join blocks Block 1: Type "You caught" Block 2: Add Score variable Type: "out of 10"		Put code stack on CASEY sprite.
	This is what the code stack looks like when it is finished.		